CLAIMPROMOTEBENCH: SOURCE-TRACEABLE REGRESSION TESTS FOR MODEL SERVICES UNDER POLICY REVISION

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Paper under double-blind review

ABSTRACT

Executable policy revisions can change the required action for an unchanged input, yet model-service tests rarely isolate policy version. We introduce CLAIMPROMOTEBENCH, which enumerates action-changing differences between two deterministic evaluators and converts them into paired route tests. It reports changed-rule and action-transition coverage, a cardinality-minimum structural suite, parser provenance, and failure-inclusive item and pair accuracy. We trace six revisions from five public projects and evaluate hosted routes on two Gatekeeper revisions. Across 144 calls, Qwen and DeepSeek Flash complete 0/36 and 21/36 raw-strict pairs. All 63 extracted actions agree with the Open Policy Agent oracle, locating the observed failures at the service or output interface. On an imbalanced evidence-state suite, Qwen reaches 113/150 pairs end to end and 113/123 conditional. An action-balanced challenge yields 48/90 and 13/90 full-contract pairs end to end for Qwen and Flash, versus 48/64 and 13/61 conditional, with `DEMOTE` recall of 4/30 and 0/30. Complete confusion matrices retain terminal failures. These results support source-traceable regression testing under finite executable policy changes. The study measures a synthetic model-service boundary; deployment frequency and naturalistic policy interpretation require field evidence.

1 INTRODUCTION

Executable decision policies evolve, and regressions can change downstream actions for unchanged inputs. Content-moderation studies document frequent updates, scarce new labels, and conflicts between historical labels and current rules (Mullick et al., 2023; Ji et al., 2026). Public software histories expose the same problem at three boundaries. Gatekeeper and Trivy changed executable policy behavior (Open Policy Agent Gatekeeper Library, 2020; 2021; Aqua Security Trivy Checks, 2024); Cosign repaired verifier semantics (Sigstore Cosign, 2022); Microsoft Agent Governance Toolkit and agentgateway repaired agent-facing policy interfaces (Microsoft Agent Governance Toolkit, 2026; Agentgateway Project, 2026). We trace six revisions across these five projects. Prior prompt-regression work fixes an application task while a hosted LLM API evolves (Ma et al., 2024). We instead fix the service route and vary the policy version.

We use *route* for the model, provider, prompt, parser, and output contract evaluated together. Our prospective test asks whether a route follows an explicit policy version and satisfies its downstream contract. A directly callable policy executor is preferable for structured inputs; here it defines the oracle for a model-mediated interface. Field studies must establish how often that interface occurs and whether testing it improves deployed systems.

Static accuracy gives an incomplete answer. A route can ignore the policy version and exploit stable field–action correlations. A response can also expose the correct action while violating a larger schema, or fail before any action is available. Valid-only scoring removes those interface failures from the denominator. Rule coverage creates another blind spot: a suite may exercise every edited rule while omitting action transitions induced by their interactions.

CLAIMPROMOTEBENCH takes two deterministic evaluators P_0, P_1, a finite context set $\mathcal{X}$ with stable identifiers, registered changed-rule witnesses, and declared coverage inventories. We execute both policies, retain contexts whose prescribed actions differ, and duplicate each context while

varying the active policy. A pair is correct only when both actions follow their respective policies. We report changed-rule and directed action-transition coverage and select a cardinality-minimum structural-coverage suite. Users remain responsible for policy parsing, grammar synthesis, rule-change discovery, and witness inference. Route evaluation separates raw-strict parsing, the registered route parser, full contract validity, and exact structured-decision agreement. Every scheduled call remains in its end-to-end denominator.

We organize the evidence in three layers. Six public histories establish policy, verifier, and adapter faults. Two Gatekeeper revisions support a 12-pair hosted study with three repetitions on two routes. A four-action SLSA-inspired migration and two evidence-state suites test transition coverage, endpoint sensitivity, and action balance. All model-facing mappings and labels are deterministic author policy mappings.

The contributions are:

1. an integrated benchmark protocol that converts finite executable policy differences into controlled policy-version pairs, relative rule/transition certificates, and a cardinality-minimum structural suite;
2. a failure-inclusive route protocol that distinguishes raw-strict, registered-parser, contract-action, and full-decision endpoints at item and pair level; and
3. a source-traceable evaluation spanning six upstream revisions in five projects, repeated hosted routes, and action-balanced evidence-state diagnostics.

The enumerator adapts established semantic-differencing and set-cover machinery. Our contribution is the test object that joins an action-changing policy intervention with paired route records, parser provenance, and a failure-inclusive endpoint.

2 Related Work

Behavioral tests, contrasts, and metamorphic relations. CheckList organizes behavioral tests by capability, and contrast sets probe local decision boundaries through controlled input changes (Ribeiro et al., 2020; Gardner et al., 2020). Metamorphic testing evaluates relations among transformed executions (Chan et al., 1998). ClaimPromoteBench uses a policy-version intervention and an exact executable oracle; its paired score detects routes whose action remains invariant across an action-changing revision.

Conformance, semantic differencing, and regression. Labelled-transition-system and finite-state-machine methods derive conformance tests from explicit behavioral models (Tretmans, 1996; Chow, 1978). Semantic differencing identifies behavioral changes between test-design models (Tzoref-Brill & Maoz, 2017). Margrave computes witnesses for access-control policy disagreement, while Cirg uses change-impact analysis to generate and reduce policy tests (Fisler et al., 2005; Martin & Xie, 2007b). Mutation testing and suite reduction supply established adequacy machinery (Martin & Xie, 2007a; Harrold et al., 1993). Prompt regression measures service evolution under a fixed application task (Ma et al., 2024). ClaimPromoteBench combines these ideas at a model-service boundary: a fixed route receives paired policy versions, rule and action-transition coverage are minimized jointly, and every service failure stays in the scored denominator. The source-traceable integration is the method contribution.

Rule and specification following. RuleArena and SpecEval test rule-governed and specification-following behavior (Zhou et al., 2025; Ahmed et al., 2025). GuideBench includes guideline updates, and TDAD studies specification evolution with hidden tests and semantic mutants (Diao et al., 2025; Rehan, 2026). Policy-to-Tests translates governance prose into executable rules (Datla et al., 2025). ClaimPromoteBench begins with two deterministic evaluators and treats policy version as the controlled coordinate. Its scope covers finite behavioral differences after policy authoring, context construction, and witness registration.

Structured output and selective prediction. Grammar-constrained decoding and incremental parsing improve syntactic validity (Scholak et al., 2021; Geng et al., 2023); JSONSchemaBench

Table 1: Positioning by controlled coordinate. The final row combines established components.

Test family	Controlled coordinate	Primary oracle or endpoint	Role in this work
Behavioral / contrast	input perturbation	labels or behavioral invariants	paired intervention
Conformance / policy diff	specification or policy	executable semantics	change oracle
Rule / specification following	textual rules	task labels	model-facing task
Structured output / abstention	schema or coverage	syntax, contract, or risk	route endpoint
Prompt regression	hosted service time	application tests	service-level regression
This work	policy version	executable action plus route contract	joint policy-to-route test

and BFCL evaluate schemas and tool arguments (Geng et al., 2025; Patil et al., 2025). Selective prediction evaluates risk under a learned reject option (Geifman & El-Yaniv, 2017; 2019). Our action set contains an explicit `HOLD` decision. Transport, syntax, schema, and semantic failures occupy a separate terminal category and remain errors end to end.

Research agents. ScienceAgentBench, CORE-Bench, and POPPER evaluate data analysis, reproducibility, and sequential falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). They evaluate broad scientific workflows; our auxiliary evidence-state suites isolate a downstream policy transition after evidence registration.

3 METHOD

3.1 FINITE POLICY-DIFFERENCE ENUMERATION

The benchmark takes deterministic evaluators P_0 and P_1, a finite set $\mathcal{X}$ of complete typed contexts with stable identifiers, registered changed-rule witnesses, and the rule inventory to cover. Let $\pi_P(c)$ denote the prescribed action. The enumerator executes both evaluators and defines

$$\Delta(P_0, P_1; \mathcal{X}) = \{c \in \mathcal{X} : \pi_{P_0}(c) \neq \pi_{P_1}(c)\}.$$

Each $c \in \Delta$ yields a pair $((P_0, c), (P_1, c))$. The context, query, and output contract are identical; only the policy source changes. This construction makes policy-version sensitivity a necessary condition for pair correctness. A deterministic route invariant to the policy returns the same action twice, so at most one member can be correct and pair accuracy is zero.

The enumerator associates every candidate with its registered changed-rule witnesses and directed action edge $(\pi_{P_0}(c), \pi_{P_1}(c))$. For suite T, adequacy is

$$A(T) = \left(\frac{|R(T)|}{|R_\Delta|}, \frac{|E(T)|}{|E_{\text{reach}}|} \right),$$

where R_Δ is the registered changed-rule inventory and E_{reach} is the action-edge set observed by exhaustive execution over $\mathcal{X}$. Both certificates are relative to the declared context set and registered witnesses. Their validity depends on context completeness and witness correctness.

Selecting the smallest structural-coverage suite with $A(T) = (1, 1)$ is a set-cover instance. We use a standard exact bit-mask dynamic program with $O(n2^{r+k})$ state updates for n candidates, r rules, and k edges. A specialized search selects one context per edge while tracking rule masks; a complete k-pair witness meets the edge lower bound and is globally minimum for the candidate pool. Appendix A gives the proof and optimality certificate.

3.2 PARSER AND CONTRACT ENDPOINTS

For scheduled item x, let $y(x) = \pi_P(c)$; let $r(x)$ be the action parsed directly from retained raw text; let $p(x)$ be the action recorded by the route-specific registered parser; let $v(x)$ indicate full

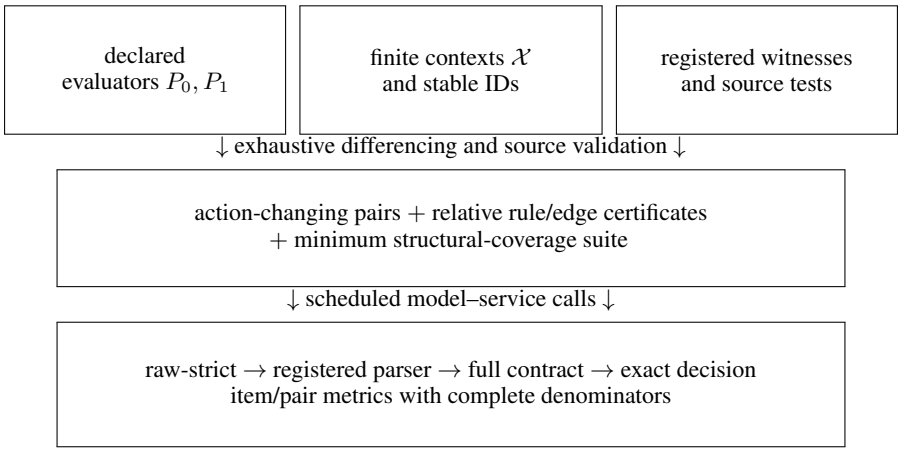


Figure 1: Benchmark and evidence flow. Declared inputs precede enumeration; source validation and suite construction finish before model calls. Each scoring layer corresponds to a distinct interface requirement.

output-contract validity; and let $d(x)$ be the parsed structured decision. The raw-strict parser accepts exactly one duplicate-free top-level JSON object and performs no fence or substring extraction, repair, coercion, or non-standard constant handling. Registered parsers retain their historical route-specific behavior. Their provenance distinguishes direct parsing from deterministic single-object extraction. No retained response contains a duplicate key. We report:

$$
\begin{aligned}
C_R(x) &= \mathbb{1}[r(x) = y(x)], && \text{raw-strict action,}\\
C_P(x) &= \mathbb{1}[p(x) = y(x)], && \text{registered-parser action,}\\
C_C(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[p(x) = y(x)], && \text{contract action,}\\
C_F(x) &= \mathbb{1}[v(x) = 1]\mathbb{1}[d(x) = d^*(x)], && \text{full decision.}
\end{aligned}
$$

The task-specific contract contains five fields for Gatekeeper and six fields for the evidence-state tasks. Pair correctness is the product of the two member indicators for the selected endpoint. End-to-end metrics divide by every scheduled item or pair. Conditional item metrics divide by items for which the selected parser or contract succeeds; conditional pair metrics require both members to succeed. Terminal failures are shown as `N` in confusion matrices, outside the policy action set.

4 Evidence and Evaluation Design

4.1 Source-traceable revisions

For Gatekeeper, we preserve the old and new Rego sources, parent and change commits, upstream test names, and license notices. Open Policy Agent 1.19.1 reproduces both changes. We construct 12 contexts around the upstream tests: six for the FlexVolume empty-list change and six for the selector-kind change. We map zero violations to `ALLOW` and one or more violations to `HOLD`, then fix the resulting 24 items before inference. Every label follows a deterministic mapping; no human annotation enters the evaluation.

The Trivy reproduction executes four upstream regression cases around the corrected inverted predicate. The Cosign reproduction exercises six cases and four verifier fault modes using the historical Go query logic and upstream tests. Cosign changes the verifier–policy interface and provides a distinct boundary-fault evidence class.

We also screened 465 public repository seeds with a fixed query and inspected 11 model–policy candidates at source level. The screen required an executable or versioned policy boundary, a finite downstream action, a historical fault, usable licensing, and source evidence. Two candidates met these criteria. For Microsoft Agent Governance Toolkit, we execute the parent and fixed Rego sources and map an undefined deny verdict to the caller fallback documented upstream. For agentgateway,

we inspect the parent implementation and run 33 Bedrock guardrail tests at the exact change commit. This fixed-query convenience sample supports case validity rather than prevalence.

Table 2: Evidence hierarchy. Structural minima are exact relative to each declared candidate set.

Instrument	Revision or instrument	Evaluation role	Size	Coverage or check
Source-anchored revisions and faults				
Gatekeeper	two upstream Rego revisions	central hosted route test	12 pairs	2 rules, 2/2 edges; min. 2
Trivy KCV0030	upstream predicate fix	policy-fault reproduction	4 cases	2/2 edges; min. 2
Cosign Rego verifier	upstream query fix	interface-fault reproduction	6 cases	2/2 edges; min. 4
AGT helper	upstream verdict fix	agent-policy interface fault	2 cases	1/1 edge; min. 1
agentgateway	Bedrock adapter fix	guardrail-interface fault	33 tests	33/33 fixed
Synthetic structural stress instrument				
SLSA-inspired	12-rule policy migration	four-action transition test	60 pairs	12 rules, 10/10 edges; min. 10
Auxiliary evidence-state diagnostics				
Imbalanced suite	evidence-state contrasts	coverage diagnostic	150 pairs	–
Balanced challenge	action-balanced contrasts	robustness diagnostic	90 pairs	–

4.2 Route studies and auxiliary instruments

The central study repeats all 24 Gatekeeper items three times on two hosted routes, for 144 scheduled calls. Each cell receives one request without client retries. We fix the schedule, prompts, route labels, endpoints, and no-retry rule before inference. Both model IDs use the same provider, base URL, OpenAI-compatible API, prompt-only output mode, and adapter settings. We report descriptive route measurements because provider-side serving internals are opaque.

The SLSA-inspired instrument defines two profiles over structured attestation fields and assigns one of four actions: `ALLOW`, `HOLD`, `DEMOTE`, or `ESCALATE` (SLSA Steering Committee, 2026; in-toto Attestation Maintainers, 2026). A rule-stratified suite covers 12 changed rules, and a graph suite covers every reachable directed action edge. We evaluate two local model routes and two registered random-forest controls. The deterministic executor defines the labels and is the *definitional policy ceiling*; learned systems are empirical controls.

The evidence-state task supplies two auxiliary suites. The imbalanced suite contains 300 items, including eight `DEMOTE` and eight `ESCALATE` labels. The balanced challenge contains 90 pairs, 180 items, and 15 factors. Its 90 action-changing members are balanced across `HOLD`, `DEMOTE`, and `ESCALATE` (30 each), and each has an `ALLOW` counterpart. We fix its cases and labels before inference. Its estimand is robustness on known cases. We include a weak multinomial Naive Bayes shortcut control for the imbalanced suite.

5 Results

5.1 Source validation and finite structural coverage

Open Policy Agent execution reproduces both Gatekeeper changes on all 12 contexts. A two-pair witness covers 2/2 changed rules and both `ALLOW`→`HOLD` and `HOLD`→`ALLOW` transitions, meeting the two-edge lower bound. Trivy reproduces two transitions in each direction around its inverted predicate, with a two-case minimum. Cosign reproduces four verifier fault modes and requires four cases to cover them.

At the Agent Governance Toolkit parent revision, 11/15 post-fix upstream tests pass; all 15 pass after the fix. Both decision cases change from an undefined deny verdict to `deny`, yielding two mapped `ALLOW`→`HOLD` transitions and a one-case minimum. For agentgateway, we inspect the parent source and execute tests at the change commit, where all 33 selected tests pass. Together these revisions instantiate policy, verifier, and guardrail-interface faults.

The four-action migration gives a separate structural result. Its 12-rule stratified suite covers 12/12 changed rules and 6/10 reachable edges. The graph suite covers 12/12 rules and 10/10 edges. An exact search selects a 10-pair, 20-call joint suite, reducing the 120-call graph suite by 83.3%. Mistral and Phi-4 score 10/60 and 2/60 pairs on the full graph. This minimum certifies coverage of the declared rules and observed edges. Stochastic fault detection and release sufficiency remain unmeasured. The registered policy-aware random-forest control reaches 107/120 items and 47/60 pairs, while the profile-blind control reaches 57/120 items and 0/60 pairs. The zero-pair result shows that paired scoring detects policy-profile invariance on this graph. The weak Naive Bayes control reaches 184/300 items and 34/150 pairs on the imbalanced suite, with balanced action recall 0.328.

5.2 Repeated route behavior on upstream revisions

The central hosted result is a gap between action correctness and interface completion. DeepSeek Flash totals 53/72 raw-strict items and 21/36 pairs; Qwen totals 10/72 items and 0/36 pairs. All 63 extracted actions match the Open Policy Agent oracle, so we observe zero semantic action errors after raw-strict extraction. Conditional item and pair accuracy is 1.0 whenever the required item or complete pair is available. Table 3 shows the three repetitions and exact-decision scores.

Terminal metadata identifies the limiting interface behavior. Qwen produces 62 Markdown-fenced objects, which the raw-strict endpoint rejects. DeepSeek Flash records 12 read timeouts, three fenced objects, three non-JSON prefixes or envelopes, and one length-terminated response. We apply no response repair or rescoring; these records remain terminal failures. Across repetitions, 13/24 DeepSeek items and 5/12 pairs pass every time; no Qwen item does.

Table 3: Gatekeeper route results by repetition. All columns are end to end. R is raw-strict action; F is exact full decision. Contract action equals R for every row.

Route	Repetition	R item	F item	R pair	F pair
Qwen	1	3/24	3/24	0/12	0/12
Qwen	2	2/24	2/24	0/12	0/12
Qwen	3	5/24	5/24	0/12	0/12
	total	10/72	10/72	0/36	0/36
DeepSeek Flash	1	15/24	15/24	6/12	6/12
DeepSeek Flash	2	19/24	18/24	7/12	6/12
DeepSeek Flash	3	19/24	19/24	8/12	8/12
	total	53/72	52/72	21/36	20/36

5.3 Endpoint sensitivity and evidence-state diagnostics

The endpoint reanalysis covers 1,320 previously scheduled items across ten route stacks. Raw-strict parsing finds 69 extractable actions outside the full six-field contract; 49 actions and 40 pairs are correct. Conversely, 452 contract-valid registered-parser actions are unavailable to raw-strict parsing, including 329 correct actions and 78 correct pairs. Most of the latter occur on historical local routes whose registered parsers used deterministic single-object extraction. Neither endpoint contains the other, and each answers a different integration question.

On the SLSA Qwen route, raw-strict action yields 120/120 items and 60/60 pairs, while the full contract yields 88/120 and 28/60. On the balanced Qwen challenge, the registered parser accepts six fenced objects, including four correct actions and two correct pairs, that raw-strict parsing rejects. Raw-strict scores are therefore 128/180 items and 46/90 pairs, compared with 132/180 and 48/90 under the full contract. The balanced Flash route changes from 105/180 and 19/90 under raw-strict parsing to 90/180 and 13/90 under the contract. Registered-parser actions and full-contract actions coincide in these records because a parser-owned decision is retained only after contract validation.

Table 4: Auxiliary item/pair metrics. All raw and contract scores are end to end; the last column is conditional on contract validity. "–" marks unavailable retained raw text.

Route	Reg. valid	Raw item/pair	Contract item/pair	Cond. item/pair
Imbalanced Qwen	272/300	–	258/300; 113/150	258/272; 113/123
Challenge Qwen	148/180	128/180; 46/90	132/180; 48/90	132/148; 48/64
Challenge Flash	146/180	105/180; 19/90	90/180; 13/90	90/146; 13/61
SLSA Qwen	88/120	120/120; 60/60	88/120; 28/60	88/88; 28/28
SLSA Flash	108/120	110/120; 51/60	108/120; 49/60	108/108; 49/49

Table 5: End-to-end contract-action recall by gold action.

Route	ALLOW	HOLD	DEMOTE	ESCALATE
Imbalanced Qwen	143/150	110/134	0/8	5/8
Challenge Qwen	79/90	26/30	4/30	23/30
Challenge Flash	68/90	9/30	0/30	13/30

Action support changes the diagnosis. Table 5 gives end-to-end contract-action recall. The imbalanced suite has eight examples for each rare action; Qwen recovers 0/8 DEMOTE and 5/8 ESCALATE. The balanced challenge increases each non-ALLOW action to 30 cases: Qwen recovers 4/30 DEMOTE, and Flash recovers 0/30. The complete confusion matrices in Appendix B retain terminal failures in every row.

6 Discussion

The regression unit crosses component boundaries. The six histories show action changes at the policy source, verifier, and guardrail adapter. Verifier and adapter changes show that faults can arise beyond the policy source; Agent Governance Toolkit and agentgateway connect this boundary to public agent or model interfaces. These cases motivate a regression unit that binds the executable policy version to the route stack, parser, and output contract. ClaimPromoteBench provides that unit.

Action choice and interface completion are separate outcomes. All 63 raw-strict Gatekeeper actions agree with the executable oracle, yet transport and serialization failures prevent many complete pairs. The endpoint reanalysis adds a second distinction: raw-strict and full-contract endpoints accept different records. A deployment-facing test should therefore name its parser, contract, and scheduled denominator.

Pairs and balanced actions expose different shortcuts. The profile-blind control completes no pair because it ignores the policy profile. The imbalanced suite conceals weak DEMOTE behavior behind abundant ALLOW and HOLD cases; the balanced challenge exposes it. Pair accuracy, action recall, and terminal columns answer different questions, so the benchmark reports all three.

The intended use is prospective. A deterministic executor remains the preferred enforcement path for fully structured inputs and defines the policy ceiling here. The Naive Bayes control supplies a weak shortcut check. ClaimPromoteBench tests model-mediated interfaces whose operational value must be measured in deployed systems.

7 Limitations

Our hosted evaluation covers two Gatekeeper revisions, one constructed context set, two model IDs, and three repetitions. The other public revisions broaden source validity without route replication. The SLSA-inspired migration broadens the action graph within the same synthetic scope. The study cannot estimate performance across policy populations, model families, or deployments.

We define the Gatekeeper, Agent Governance Toolkit, SLSA, and evidence-state action mappings. No human annotation enters the evaluation. The fixed-query screen covered 465 repository seeds and inspected 11 candidates, making its two qualifying cases a convenience sample of boundary faults. No retained external case tests a model interpreting a changed executable policy; naturalistic validity requires such a case.

Each coverage claim is conditional on a finite context set and registered witnesses. Missing contexts can hide behavior, and incorrect witnesses invalidate rule coverage. Minimum-suite optimality covers the declared rule and edge universe; stochastic fault-detection power and release sufficiency remain unmeasured. Trivy requires a documented compatibility helper for historical Rego, while Cosign evaluates verifier logic rather than policy-source differencing.

Hosted results combine model weights, prompts, constraints, provider state, and time. Both central model IDs share one provider, base URL, API, output mode, and adapter configuration, while serving internals are opaque. The historical Flash configuration for the imbalanced suite used an obsolete service interface: all 300 requests returned HTTP 401 before model output. Later Flash calls show that their newer paths execute. Because the service and output stack changed, the 401 lane provides no causal comparison.

We fixed the balanced challenge before inference, and its estimand is robustness on known cases. The registered parser accepts six fenced Qwen outputs through deterministic single-object extraction; raw-strict parsing rejects them. Policy retrieval, ambiguous source text, evidence extraction, and downstream utility fall outside the evaluation.

8 Conclusion

ClaimPromoteBench turns executable policy changes into paired route records with rule and transition coverage, a cardinality-minimum structural suite, parser provenance, and failure-inclusive endpoints. Public revision histories establish the target fault classes; hosted and action-balanced tests show how interface completion and action recall can fail independently. A deterministic executor remains the definitional policy ceiling. The next discriminating experiment is a source-traceable deployed route in which a model interprets a changed policy across independent service stacks.

9 Reproducibility Statement

The released artifact binds policies, datasets, prompts, source tests, route configurations, and parser implementations. Its deterministic replay recomputes source bindings, coverage, minimum suites, endpoint metrics, action recalls, and confusion matrices without reissuing hosted calls. The anonymous bundle removes private service metadata and hidden reasoning.

10 AI Assistance Disclosure

Generative AI assisted literature search, code development, analysis checks, and editing. The author selected the research question and protocol, verified sources, results, and artifacts, and assumes responsibility for the paper.

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A Enumerator and Reducer Guarantees

Minimum-suite hardness. Take a Set Cover instance with universe U and subsets $S_1, \ldots, S_n$. Create one required rule for each element of U and one candidate context for each S_i whose rule-witness set is S_i. Assign every candidate the same required action edge. Any nonempty selection covers that edge, and it covers all required rules exactly when the corresponding subsets cover U. A minimum joint suite therefore gives a minimum set cover.

Exact dynamic program. Index the $r + k$ required rules and edges and encode coverage as a bit mask. After processing the first j candidates, retain for every reachable mask the shortest subset, breaking ties by context identifier. Excluding candidate $j + 1$ preserves each state; including it unions its coverage mask with each prior mask. Induction on j proves that the final full-mask representative is cardinality-minimum. The implementation matches exhaustive enumeration on 64 deterministic small candidate pools.

Edge lower bound. Each pair realizes one directed edge. Covering k edges requires at least k pairs. If the specialized search selects one candidate per edge and also covers every rule, its k-pair witness meets the lower bound. This certifies the two-pair Gatekeeper suite and the ten-pair SLSA-inspired suite relative to their declared candidate pools.

B Complete Action-Level Results

Table 6: End-to-end action recall for every route summarized in the main text. "–" marks actions outside a task's label set.

Configuration and endpoint	ALLOW	HOLD	DEMOTE	ESCALATE
Gatekeeper Qwen, raw-strict	4/36	6/36	–	–
Gatekeeper Flash, raw-strict	32/36	21/36	–	–
Imbalanced Qwen, contract	143/150	110/134	0/8	5/8
Challenge Qwen, raw-strict	77/90	26/30	3/30	22/30
Challenge Qwen, contract	79/90	26/30	4/30	23/30
Challenge Flash, raw-strict	68/90	10/30	2/30	25/30
Challenge Flash, contract	68/90	9/30	0/30	13/30
SLSA Qwen, raw-strict	60/60	20/20	20/20	20/20
SLSA Qwen, contract	60/60	20/20	1/20	7/20
SLSA Flash, raw-strict	59/60	17/20	17/20	17/20
SLSA Flash, contract	59/60	17/20	15/20	17/20
Graph Mistral, contract	9/36	18/24	0/24	13/36
Graph Phi-4, contract	21/36	5/24	0/24	1/36

Table 7: Gatekeeper aggregate raw-strict confusion counts over three repetitions (left) and SLSA-inspired transition-graph contract-action confusion counts (right). N marks the endpoint's terminal failure category.

Gatekeeper

Route	Gold	ALLOW	HOLD	N
Qwen	ALLOW	4	0	32
Qwen	HOLD	0	6	30
Flash	ALLOW	32	0	4
Flash	HOLD	0	21	15

SLSA-inspired graph

Route	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Mistral	ALLOW	9	8	0	9	10
Mistral	HOLD	2	18	0	4	0
Mistral	DEMOTE	3	0	0	0	21
Mistral	ESCALATE	2	3	0	13	18
Phi-4	ALLOW	21	0	0	0	15
Phi-4	HOLD	1	5	0	0	18
Phi-4	DEMOTE	3	12	0	0	9
Phi-4	ESCALATE	0	0	0	1	35

Table 8: Imbalanced-suite and balanced-challenge full-contract confusion counts. Columns are terminal predicted actions; N is a contract failure.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Imbalanced Qwen	ALLOW	143	0	0	0	7
Imbalanced Qwen	HOLD	4	110	0	2	18
Imbalanced Qwen	DEMOTE	8	0	0	0	0
Imbalanced Qwen	ESCALATE	0	0	0	5	3
Challenge Qwen	ALLOW	79	0	0	0	11
Challenge Qwen	HOLD	0	26	0	0	4
Challenge Qwen	DEMOTE	8	4	4	4	10
Challenge Qwen	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	0	8	10
Challenge Flash	HOLD	13	9	0	0	8
Challenge Flash	DEMOTE	25	3	0	0	2
Challenge Flash	ESCALATE	0	3	0	13	14

Table 9: Balanced-challenge raw-strict action confusion counts. N marks a terminal extraction failure.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Challenge Qwen	ALLOW	77	0	0	0	13
Challenge Qwen	HOLD	0	26	1	0	3
Challenge Qwen	DEMOTE	7	3	3	4	13
Challenge Qwen	ESCALATE	0	0	0	22	8
Challenge Flash	ALLOW	68	4	5	13	0
Challenge Flash	HOLD	13	10	7	0	0
Challenge Flash	DEMOTE	25	3	2	0	0
Challenge Flash	ESCALATE	0	3	2	25	0

Table 10: SLSA-inspired raw-strict and contract-action confusion counts. N is the endpoint's terminal failure category.

Route	Endpoint	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	N
Qwen	raw-strict	ALLOW	60	0	0	0	0
Qwen	raw-strict	HOLD	0	20	0	0	0
Qwen	raw-strict	DEMOTE	0	0	20	0	0
Qwen	raw-strict	ESCALATE	0	0	0	20	0
Qwen	contract	ALLOW	60	0	0	0	0
Qwen	contract	HOLD	0	20	0	0	0
Qwen	contract	DEMOTE	0	0	1	0	19
Qwen	contract	ESCALATE	0	0	0	7	13
Flash	raw-strict	ALLOW	59	0	0	0	1
Flash	raw-strict	HOLD	0	17	0	0	3
Flash	raw-strict	DEMOTE	0	0	17	0	3
Flash	raw-strict	ESCALATE	0	0	0	17	3
Flash	contract	ALLOW	59	0	0	0	1
Flash	contract	HOLD	0	17	0	0	3
Flash	contract	DEMOTE	0	0	15	0	5
Flash	contract	ESCALATE	0	0	0	17	3